

ILYA CHEKHOV

Senior Python Backend Engineer / Tech Lead

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Open to fully remote roles worldwide

Professional Summary

Senior Python Backend Engineer / Tech Lead with 6+ years of commercial experience designing and scaling high-load backend systems and microservice architectures using Python, Django, FastAPI, asyncio, PostgreSQL, and RabbitMQ. Currently building the microservices backend of a food-media and e-commerce platform at X5 Group, Russia's largest food retailer, with FastAPI, PostgreSQL, RabbitMQ, and Kubernetes. Experienced in technical leadership: architecture design, technical specifications, production ownership, code review, and mentoring. Worked on a platform serving 630,000+ users and delivered up to 60% lower database load, up to 50% lower RabbitMQ memory consumption, and 20% fewer message-processing failures. Open to fully remote Senior / Lead roles.

Technical Skills

Languages & Frameworks: Python, Django, Django REST Framework (DRF), FastAPI, asyncio, Celery, SQLAlchemy, Alembic

Data & Messaging: PostgreSQL, ClickHouse, Redis, RabbitMQ, Apache Kafka, gRPC

Infrastructure & DevOps: Docker, Kubernetes (K8s), Helm, Ansible, GitLab CI/CD, Linux

Observability & Security: Grafana, Prometheus, Keycloak, HashiCorp Vault

Architecture & Practices: Microservices architecture, REST API design, system design, distributed systems, Pytest, LLM integration

Engineering Leadership: Technical leadership, architecture design, code review, mentoring, onboarding, technical specifications, production incident management

Work Experience

Lead Python Developer — X5 Group

Remote | Jun 2025 – Present

Food-media and e-commerce platform of X5 Group, Russia's largest food retailer (Pyaterochka, Perekrestok); microservices backend of ~20 Python/FastAPI services.

- Introduced RabbitMQ as the platform-wide event-driven messaging layer, standardizing asynchronous communication between microservices and establishing shared engineering practices for the team; designed and deployed the production message-broker infrastructure on Kubernetes with high availability and monitoring.
- Designed and delivered the platform's video-processing service from scratch — upload, asynchronous transcoding, processing lifecycle, and adaptive streaming — enabling a new product capability, using S3-compatible storage, FFmpeg, and HLS.
- Built the content recommendation backend with FastAPI and Keycloak-authenticated service integration, implementing a cached cold-start strategy that eliminated per-request database queries for new users.
- Built a reliable master-data synchronization pipeline that automates product-category data flow from the retailer's master-data system into PostgreSQL across three retail chains, with idempotent processing and transactional consistency.
- Shortened the CI feedback loop for the team by parallelizing the test suite and introducing incremental test selection (pytest-xdist, pytest-testmon), with enforced coverage gates in GitLab CI.
- Owned Helm-based deployment of ~20 microservices and 26 scheduled jobs across production and 18 staging environments, managing environment configuration as code and reducing operational overhead.

Lead Python Developer / Team Lead — Skillbox

Remote | Dec 2022 – May 2025

One of Russia's largest EdTech platforms (#1 in continuing-professional-education training quality, Smart Ranking); 630,000+ users.

- Led architecture design, technical specifications, release management, and production monitoring for the core learning platform and its supporting microservices serving 630,000+ users.
- Optimized PostgreSQL performance through query, schema, and indexing improvements, cutting database load by up to 60%.
- Reduced RabbitMQ memory consumption by up to 50% and message-worker failures by 20% through queue architecture and consumer tuning.
- Delivered CRM (S2) and CDP (Mindbox) integrations and migrated transactional and marketing email delivery to a new SMTP provider, unifying customer data and communications across the platform.
- Redesigned the core lessons data model (foreign-key to many-to-many), unlocking flexible course structures and improving query performance.
- Mentored and onboarded engineers; established code-review standards and engineering best practices across the team.

Senior Python Developer — Playrix

Remote | Jul 2021 – Dec 2022

Global mobile-gaming company (Gardenscapes, Homescapes).

- Built and maintained internal enterprise services: an employee self-service portal, electronic document management (EDM) integration, and a game-data management platform.
- Automated internal business processes, reducing manual operational workload across departments.

Python Developer — Intekh

Remote | May 2021 – Jul 2021

- Developed backend microservices for a self-service warehouse platform: mobile-app smart-lock access, automated payment deduction, and inventory tracking.

Python Developer — Tensor

Yaroslavl, Russia | Dec 2019 – May 2021

Enterprise software vendor (accounting and reporting systems).

- Developed and maintained a management-accounting system and automated tax-reporting workflows for enterprise clients.

Education

Specialist Degree (5-year university degree), Computer Security
Saint Petersburg Electrotechnical University “LETI”, Saint Petersburg, Russia

2020

Languages

Russian — native | **English** — professional working proficiency (CEFR B2)